

Algunas Distribuciones Continuas

Nombre	Parámetros	fdp/Fda	Media	Varianza	fgm
Beta $B(\alpha, \beta)$	$\alpha, \beta > 0$	$\frac{x^{\alpha-1}(1-x)^{\beta-1}}{B(\alpha, \beta)}$ $x \in [0, 1]$	$\frac{\alpha}{\alpha+\beta}$	$\frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$	
Chi-cuadrado $\chi^2_{(\nu)}$	$\nu \in \mathbb{N}^*$	$\frac{x^{\frac{\nu}{2}-1}e^{-\frac{x}{2}}}{2^{\frac{\nu}{2}}\Gamma(\frac{\nu}{2})}$	k	$2k$	$(1-2t)^{-\frac{k}{2}}$ $t < \frac{1}{2}$
Exponencial $Exp(\lambda)$	$\lambda > 0$	$f(x) = \lambda e^{-\lambda x}, x > 0$ $F(x) = 1 - e^{-\lambda x}$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$	$\frac{\lambda}{\lambda - t}$ $t < \lambda$
$F_{(\nu_1, \nu_2)}$	$\nu_1, \nu_2 > 0$	$\frac{\sqrt{\frac{(\nu_1 x)^{\nu_1} (\nu_2)^{\nu_2}}{(\nu_1 x + \nu_2)^{\nu_1 + \nu_2}}}}{x B(\frac{\nu_1}{2}, \frac{\nu_2}{2})}$	$\frac{\nu_2}{\nu_2 - 2}$ $\nu_2 > 2$	$\frac{2(\nu_2)^2(\nu_1 + \nu_2 - 2)}{\nu_1(\nu_2 - 2)^2(\nu_2 - 4)}$	

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Gama $\Gamma(\alpha, \theta)$	$\alpha, \theta > 0$	$\frac{x^{\alpha-1} e^{-\frac{x}{\theta}}}{\Gamma(\alpha)\theta^\alpha}$ $x > 0$	$\alpha\theta$	$\alpha\theta^2$	$(1 - \theta t)^{-\alpha}$ $t < \frac{1}{\theta}$
Normal $N(\mu, \sigma)$	$\mu \in \mathbb{R}$ $\sigma > 0$	$\frac{1}{\sqrt{2\pi}\sigma^2} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$ $x \in \mathbb{R}$	μ	σ^2	$e^{\mu t + \sigma^2 \frac{t^2}{2}}$
Student $t_{(\nu)}$	$\nu > 0$	$\frac{1}{\sqrt{\nu} B(\frac{1}{2}, \frac{\nu}{2})} \left(1 + \frac{x^2}{\nu}\right)^{\frac{\nu+1}{2}}$	0	$\frac{\nu}{\nu-2}$ $\nu > 2$ ∞ si $1 < \nu \leq 2$	
Uniforme $U(a, b)$	$0 < a < b$	$f(x) = \frac{1}{b-a} \setminus F(x) = \frac{x-a}{b-a}$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$	$\begin{cases} \frac{e^{tb}-e^{ta}}{t(b-a)} & \text{si } t \neq 0 \\ 1 & \text{si } t = 0 \end{cases}$